



CPSC - 6300 - 500
Applied Data Science
Spring 2026

Instructor Information



Dr. Tim Ransom
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Course Description

Being able to extract knowledge from large, complex data sets is one of the most critical skills in today's data-driven world. This course provides an introduction to fundamental concepts and techniques of Data Science. Students will learn to combine tools and methods from computer science, statistics, data visualization, and the social sciences to extract knowledge from data. Concepts taught in the course will be illustrated with case studies drawn from fields such as business, public health, and the social sciences. This class focuses on teaching library (e.g, Pandas) based data analysis and model development.

Course Learning Objectives

At the end of the semester, all students taking this course should be able to do the following:

- Use Python and other tools to scrape, clean and process data

- Use visualization techniques to explore data and communicate results
- Implement basic machine learning algorithms on data, and interpret their output for inference and prediction
- Examine and compare the performance of different data science models, and select a best performing model for an application
- Design a study to investigate the applicability and performance of various data science models for a given data set
- Report the results of a data analysis effectively (visually, verbally, and in writing), and defend the validity of results with respect to model performance and selected methods
- Examine ethical issues in data science

Course Schedule

1. Course Outline
2. Data, Statistics, & Visualization
3. Regressions
 1. Linear
 2. Logistic
4. Model Selection
5. Clustering
6. Outlier Detection
7. Association Rule Mining
8. Support Vector Machines
9. Perceptrons
10. Neural Networks
 1. Architectures
 2. Optimization & Regularization
11. Convolutional Neural Networks
12. Ethics of Data Science

Course Information

Course Modality: Online

Course Grading Scheme and Grading Policies

Final grades will be based on the final project, labs, quizzes, and homework assignments. Students will need to complete a final project, which includes analyzing an unknown data set and submitting a report. All graded components and project goals are expected to be of a high standard.

Quizzes	25%
Labs & Assignments	50%
Final Project & Checkpoints	25%
Total	100%

Letter grades in this class will use the following table to convert from numerical score to letter grade:

letter grade	undergraduate sections	graduate sections
A	[90,100)	[95,100)
A-		[90,95)
B+		[86,90)
B	[80,90)	[83,86)
B-		[80,83)
C+		[76,80)
C	[70,80)	[73,76)
C-		[70,73)
D+		[66,70)

letter grade	undergraduate sections	graduate sections
D	[60,70)	[63,66)
D-		[60,63)
F	[0,60)	[0,60)

Office Hours

Synchronous, online office hours will be held weekly. The exact time will be determined near the beginning of the semester.

Course Absence/Attendance Policy

As an online, asynchronous class, there is no required attendance policy. It is the responsibility of the student to engage with the material and contact the instructional staff when needed.

AI Policy

The use of generative AI technology is by default disallowed in this class. Specific instances of it's permitted use will be indicated on a per assignment bases through the LMS (canvas). Note that this does not restrict the use of grammar and spell-checking tools such as Grammarly or Microsoft Word's spell-checker. Instances of suspected unethical use of computing in this class will proceed with the Clemson University official policy and may result in academic misconduct charges.

Unacceptable uses of artificial intelligence in this class include:

- Drafting text
- Editing text
- "Brainstorming" ideas
- Outlining

CECAS Resources Geared for STEM Majors

PEER&WISE offers free tutoring and academic support to all students in the College of Engineering, Computing, and Applied Sciences (CECAS), across all departments in Engineering and Computing. Our tutoring program focuses on high-demand and historically challenging courses, and all tutors are CRLA Level 1 certified to ensure high-quality academic support. Whether you're preparing for exams,

reviewing complex concepts, or improving your study strategies, our trained team is here to help you succeed.

To learn more and access tutoring resources,

visit: <https://www.clemson.edu/cecas/departments/peer-wise/students/resources/>

For additional information or questions please email peerwisetutoring@clemson.edu or to schedule an appointment please email for scheduling information via CU Navigate.

Spring 2026 Academic Calendar

January

- Wednesday, January 7: Classes Begin
- Tuesday, January 13: Last day to register or add a class or declare Audit
- Monday, January 19: Martin Luther King Jr. holiday (University closed, classes do not meet)
- Wednesday, January 21: Last day to drop a class or withdraw from the University without a W grade
- Wednesday, January 28: Last day to apply for May graduation

February

- Friday, February 27: Last day to issue midterm evaluations

March

- Friday, March 13: Last day to drop a class or withdraw from the University without final grades
- Mon-Fri, March 16-20: Spring break (University Open, classes do not meet)

April

- Monday, April 6: Registration for summer and fall terms begins
- Thurs-Fri, April 23-24: Classes meet; exams permitted in labs and one-hour courses only
- Monday, April 27 - Friday May 1: Examinations

May

- Monday, May 4: 9:00 AM - Deadline to submit candidate grades
- Wednesday, May 6: 9:00 AM - Deadline to submit other grades

*Academic Calendar dates are subject to change as conditions warrant.

University Policies

You can find information about campus wide policies, including student affairs information and accessibility services, by clicking on [Clemson's University Policies Page](#). There, you can find information about academic integrity, access and equity (including student accessibility and Title IX info), student financial services, emergency planning, and more. My expectation is that you will review these policies carefully and be responsible for them this semester.

Student Accessibility Services

Clemson University values the diversity of our student body as a strength and a critical component of our dynamic community. Students with disabilities or temporary injuries/conditions may require accommodations due to barriers in the structure of facilities, course design, technology used for curricular purposes, or other campus resources. Students who experience a barrier to full access to this class should let the instructor know and make an appointment to meet with a staff member in [Student Accessibility Services](#) as soon as possible. You can make an appointment by calling 864-656-6848, by emailing CUSAS@clemson.edu, or by visiting Suite 239 in the Academic Success Center building. Appointments are strongly encouraged – drop-ins will be seen, if at all possible, but there could be a significant wait due to scheduled appointments. Students who have accommodations are strongly encouraged to [request, obtain, and send these](#) to their instructors via SAS as early in the semester as possible so that accommodations can be made in a timely manner. It is the student's responsibility to follow this process each semester.

Title IX

The Clemson University Title IX statement: Clemson University is committed to creating and continuously fostering a caring community based on the core values of integrity, honesty and respect. Sexual discrimination, which includes sexual harassment, sexual violence, stalking and domestic and/or relationship violence, is unacceptable and has no place in Clemson's community. Consistent with its Title IX obligation, the University prohibits discrimination, including sexual and gender-based harassment and violence, in all its programs and activities, including academics, employment, athletics, and other extracurricular activities. This [Title IX policy](#) is available online. Katherine Weathers is the Clemson University Title IX Coordinator and VP of Inclusive Excellence. She can be reached at (864) 656-3413 or via email at kweath3@clemson.edu. Remember, email is not a fully secured method of communication and should not be used to discuss Title IX issues.

Academic Integrity

As members of the Clemson University community, we have inherited Thomas Green Clemson's vision of this institution as a "high seminary of learning." Fundamental to this vision is a mutual commitment to truthfulness, honor, and responsibility, without which we cannot earn the trust and respect of others. Furthermore, we recognize that academic dishonesty detracts from the value of a Clemson degree. Therefore, we shall not tolerate lying, cheating, or stealing in any form. Using materials generated using artificial intelligence (AI) that are turned in without attribution is considered plagiarism.

All infractions of academic dishonesty by undergraduates must be reported to Undergraduate Studies for resolution through that office. In cases of plagiarism instructors may use the Plagiarism Resolution Form. See the following resources:

- [Undergraduate Academic Integrity Policy](#)
- [Graduate Student Handbook](#)

Online Testing

By enrolling in a course with online assessments, the student agrees and consents to the use of an online test proctoring service or software program. The student also agrees to allowing the student, their activity, and surrounding workspace to be recorded by video and audio and then analyzed by the test proctoring system, the course instructor, and others at Clemson University. For more information see the Academic Regulations section of the Undergraduate Catalog.

Tech Support

If you have trouble with Canvas or another university system, check here first: [Clemson System Status](#). There, you can see if there is a current issue and when it might be resolved. If there's no current issue listed, try logging out and quitting your browser before trying again. You can also look in the [Canvas Help Guides](#) for more information.

The CCIT Support Center at Clemson offers a wide range of support options and hardware repair with several contact methods to help you answer your questions as quickly as possible:

- Phone: (864) 656-3494
- Email: ITHelp@clemson.edu
- Entire catalog of IT services, including an online chat and the Knowledge Base at: [IT Help and Support](#)

Officially, Canvas supports all the major web browsers: Chrome, Firefox, Edge, and Safari. However, Safari is not fully compatible; images often do not show up in Safari, so Clemson Online recommends

against using it with Canvas.

Canvas also has mobile apps available for teachers, students, and parents, all of which can be found in your mobile device's app store. These apps are convenient, but do not necessarily display or connect to every component of a Canvas site. They are good for assignment reminders, announcements, and messaging, but not as good for reading materials, taking exams, or building elements.

Emergency Preparedness Statement

[Emergency procedures](#) can be found on the Clemson University Emergency Management website and in the purple-covered flipbooks located in offices and classrooms through campus. Students and employees should ensure they are signed up to receive [CU Alert](#) text messages, familiarize themselves with emergency procedures and follow all directions from emergency responders in the event of an emergency. More information about safety resources available at Clemson University can be found at Clemson.edu/CUSafety.

Clemson Libraries

Do you need academic sources but don't know where to start?

Don't spend hours searching on your own -- ask a librarian! The Libraries provide time saving [online guides for subject areas](#) that can help you find articles, databases, books and more. Personalized research assistance with [librarians who specialize in subject areas](#) is available by appointment. Help is available in person at each of our many [locations](#) and you can also chat with a librarian live from our [website](#), or text [864.762.4884](#).

Do you need to verify a book, article, or data set suggested by AI to support your research or paper?

You can check out the [online catalog](#), use the [Libraries AI Research Assistant](#), use verified AI research tools embedded in databases like [JSTOR](#) and [EBSCO](#), or chat with a librarian for quick help. There is also an online guide about [AI Tools for Research](#) and a [10 Step AI Challenge](#) that you can explore to find out more about AI tools.

Not sure how to cite a source properly in your bibliography?

There are [online guides](#) to help you with any citation style as well as citation management tools available for Clemson students: [RefWorks](#) & [Endnote](#). There are also [interactive online courses](#) with tips to help you understand and avoid plagiarism.

Working on a creative project like creating a video or 3D object?

For assistance with digital and hands-on creative projects, the [Adobe Studio and Makerspace](#) is located on the 5th floor of Cooper Library and is staffed with experts who can help turn your creative ideas into reality. You can even [check out many different kinds of equipment](#) like high resolution cameras, green screens, and projectors from our library services desk in Cooper Library.

Need help making a chart or graph to help visualize your data?

The [Data Visualization Lab](#) on the 4th floor of Cooper provides support for data visualization, data analysis, and digital research methods.

No matter what your research needs, check out what the Clemson Libraries have to offer! www.libraries.clemson.edu

Online Course Etiquette ("Netiquette")

Even if you have experience communicating in an online environment, online courses may require an extra effort to be civil. An online classroom is still a classroom and we should all behave professionally while in it. Please read and follow the guidelines below.

While in our class, please remember:

- Be respectful. Consider the person on the receiving end of your comments. If you were sitting in front of that person, would you say the same thing to their face?
- Humor and sarcasm don't always translate well in online discussions. Always consider your tone and whether your words can be clearly understood by your classmates.
- Only reveal information that you are comfortable sharing with your classmates.
- Please spell-check your responses; you can even compose your work in a different program (such as Word), and then paste it into the online discussion.
- Posts don't have to be perfect, but they should be as clear and readable as you can make them.
- The law still applies on the Internet. Do not commit illegal acts online, such as libeling or slandering others, and do not joke about committing illegal acts.

Please do not:

- Use offensive language, even in jest. Any comments that can be construed as racist, sexist, etc., may be removed.
- Avoid using caps lock to post messages or responses to posts. All-caps is considered SHOUTING AND IS VERY RUDE.
- Post large, unbroken blocks of text. These can be difficult to read. Try to break up your post into paragraphs.
- Post a message more than once; if you are having trouble seeing or submitting posts, contact me and we will try to fix it.
- Re-post or forward any communication that is sent to you unless you receive explicit permission to do so. Respect the response you get if your request is denied.

Professional Communication

We will use Canvas (<http://www.clemson.edu/canvas/>) in this class. The weekly course schedule will be conveyed via Canvas on the home page. It is your responsibility to check Canvas and your Clemson email at least once a day.

All email to the instructor and TA must come from your Clemson email address. Please allow 1-2 business days for the instructor and/or TA to respond to your email. Professional emails will receive priority over unprofessional, consult your preferred search engine for tips [such as these](#) about how to write a polite and professional message.

We will not correspond with your parents or guardians about your grades.

Changes to Syllabus

Lecture topics and assignments are subject to change. The course syllabus is a general plan for the course; deviations to the class may be necessary and will be announced to class by the instructor.